MIS41110 Practical Sheet, Week 1

Covers the Week 1 lecture (types, math, strings, variables) · **Released** with the Week 1 slides · **Worked** Monday of Week 2 · **Solutions** that afternoon

*Work the exercises in order; your tutors release the solutions after the day's last session. Welcome to **Farpoint Station**, Antarctica: for the rest of the term, the practicals run its penguin counts, supplies, and sensors.*

Exercise 0: the setup gate (do this first)

Everything this term happens in **Python 3 + a terminal + VS Code**. Before any exercise, prove your machine is ready:

1. Open a terminal and run `python3` (Windows: `python`). You should see the `>>>` prompt; type `2 + 2`, get `4`, then leave with `quit()`.
2. In VS Code, create a file `hello_farpoint.py` containing one line that prints `Farpoint Station: 44 crew, 4 penguin species, 1 very long night.` Run it **from the terminal**: `python3 hello_farpoint.py`. The terminal must be *in the folder where the file lives*: easiest is VS Code's own terminal (menu: Terminal, then New Terminal), which opens right next to your file. If Python says `can't open file`, you're in the wrong folder: `cd` to it (one new terminal word: change directory), or reopen the VS Code terminal.
3. The line below has one bug. Paste it into your file, run it, read the error, fix it:

```
print("Radio check: all stations, please respond)
```

Show a tutor the working run; if there's a queue, carry on to Exercise 1 and wave one over when free. (No laptop yet? The unsupported browser stopgaps from the overview cover Exercises 1 to 4 today, though this exercise needs the real terminal. Fix the setup this week.)

Exercise 1: supply-crate arithmetic

A resupply drop leaves `a = 17` fuel drums and `b = 5` sledges. Write a program that stores the two values in variables and prints, using them (short names are fine here: this drill prints algebra):

```
Value a is <class 'int'>
Value b is <class 'int'>
17 + 5 = 22
17 - 5 = 12
17 * 5 = 85
17 / 5 = 3.4
17 % 5 = 2
17 ** 5 = 1419857
```

(a) Change the two stored values; every printed line must update without touching the `print` statements. (Keep `b` nonzero; division by zero is a different lesson.)

(b) One of the six arithmetic lines prints a different *type* than the others (the first two lines print a `type`, not a value). Which one, and why?

Exercise 2: the station door sign

Using **one** `print()` call and one string (escapes allowed), produce the sign:

```
==============================
   FARPOINT STATION
   "South of everywhere"
   Crew: 44   Winter: yes
==============================
```

(a) The `=` lines must come from string multiplication (`"=" * 30`), not typing thirty characters.

Exercise 3: temperature math

The night shift logs three readings: `-17.1`, `-17.1`, `-17.1`.

1. Print their sum. Look closely at the output.
2. In a comment in your file, explain in one sentence *why* the sum isn't exactly `-51.3` (the week-1 wall).
3. Print the average of the three readings, and the value of the formula `(9 / 5) * (-17.1) + 32` (the reading in Fahrenheit). Notice the average comes out clean: float errors can cancel too, as the lecture said.

Exercise 4: predict, then run

For each line, write your prediction as a comment first, then run and compare:

```python
print(2 + 3 * 4)
print((2 + 3) * 4)
print(3 * 2 ** 2)
print(7 // 2)
print(7 % 2)
print(True + True)
print(str(3) + str(0.5))
print(int("13") + 5)
print("13" + str(5))
```

(a) Two of the lines produce a *string* rather than a number. Which two?

Meaningful variable names, a comment where a line needs one, and run your file after every change.